

# Rebecca Prezgay

**Graduate Student**

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## Education

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**Master's** California State Polytechnic University, Pomona August 2025- Present  
Master of Science, Biology

**BS** California State University, Long Beach December 2023  
Bachelor of Science: Marine Biology

Fullerton College, Biology (Transfer) 2019-2021

## Research Experience

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**Graduate Research**, California State Polytechnic University, Pomona August 2025-Present  
Advisor: Andrea Bonisoli-Alquati, PhD

- *Thesis Research*- Pesticides in the Marine Environment. Quantifying new and phased-out pesticides in Red and Purple Sea Urchins.
- *Lab Training*- DNA extractions, PCR, gel electrophoresis

**Research Assistant-Shark Lab**, CSULB August 2021-May 2025  
Supervisor: Christopher G. Lowe, PhD

- Wetland and Beach Seines, Otter Trawls, and Longline fishing
- Measurements of elasmobranchs, taking samples (blood, muscle, and fin clips), and releasing animals with an internal or external tag.
- *Image Analysis*- measurements of organisms from drone (UAV) footage using ImageJ
- *Triangulation Data Analysis*- using receiver array positional data to triangulate the exact location of a shark based on multiple receiver detections using multilateration in Python
- Maintenance of Innovasea acoustic receivers, including battery changes
- Elasmobranch Necropsies and Preservation

**Research Assistant- Institute for Integrated Research on Materials, Environment and Society (IIRMES)**, CSULB January 2024-May 2025  
Supervisor: Varenka Lorenzi, PhD

- Overview- Collecting muscle and liver samples from elasmobranchs in suspected contaminated and uncontaminated areas in Southern California to quantify levels of organochlorine pesticides (DDTs and chlordanes) and PCBs
- Lab Skills - solvent extraction of elasmobranch muscle and liver tissue using Soxhlet extractors and processing samples for Gas Chromatography-Mass Spectrometry (GC-MS) analysis, including concentration of samples by rotary evaporators, and cleaning with alumina-silica columns

## Scientific Publications

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Rex, P.T., Abbott, K.J., Prezgay, R.E., Lowe, C.G. 2024. The Effects of Depth and Altitude on Image-Based Shark Size Measurements Using UAV Surveillance. *Drones*, 8, 547.  
<https://doi.org/10.3390/drones8100547>

## Conference Presentations & Special Talks

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(Poster) **Rebecca Prezgay**, Varenka Lorenzi, Jeremy T. Claisse, Andrea Bonisoli-Alquati. Exposure to Legacy and Current Use Pesticides in Sea Urchins of Aquaculture and Fisheries Interest – SoCal SETAC 2026. San Pedro, California. April 26-27, 2026.

(Poster) Victoria Gronwald and **Rebecca Prezgay**. The Secret Lives of Panamanian Stingrays - College of Science Symposium 2026. Pomona, California. April 24, 2026.

(Poster) **Rebecca Prezgay**, Varenka Lorenzi, Jeremy T. Claisse, Andrea Bonisoli-Alquati. Exposure to Legacy and Current Use Pesticides in Sea Urchins of Aquaculture and Fisheries Interest– College of Science Symposium 2026. Pomona, California. April 24, 2026.

(Presentation) **Rebecca Prezgay**, Jeremy T. Claisse, Andrea Bonisoli-Alquati. Exposure to Current-Use and Legacy Pesticides in Sea Urchins - Agriculture Research Institute (ARI) Next-Gen. 2025-2026 Workshop Four. April 17, 2026, via Zoom.

(Presentation) **Rebecca Prezgay**, Jeremy T. Claisse, Andrea Bonisoli-Alquati. Exposure to Current-Use and Legacy Pesticides in Sea Urchin. Agriculture Research Institute (ARI) Next-Gen 2025-2026 Workshop Three. February 6, 2026, via Zoom.

(Presentation) **Rebecca Prezgay**, Jeremy T. Claisse, Andrea Bonisoli-Alquati. Exposure to Legacy and Current Use Pesticides in Marine Invertebrate and Kelp Species of Aquaculture and Fisheries Interest The Southern California Regional Chapter of the Society of Environmental

Toxicology and Chemistry (SoCal SETAC) 2026 Winter Dinner Meeting Presentations. January 27, 2026, Santa Monica, CA.

(Poster) **Rebecca Prezgay**, Luci Herrera-Lopez, Vanessa VanDeuson, Ryan Freedman, Christopher G. Lowe, Kady Lyons, Varenka Lorenzi. *Implementation of Liver Biopsies to Quantify Organochlorine Contaminants in Elasmobranchs* – SoCal SETAC 2025. San Diego, California. April 28-29, 2025.

(Special Talk) **Rebecca Prezgay**. Ocean Contaminants and Safe Fish Consumption. Coffee and Conversations with OC Habitats. December 4, 2024

(Poster) **Rebecca Prezgay**, Luci Herrera-Lopez, Vanessa VanDeuson, Ryan Freedman, Varenka Lorenzi, Kady Lyons, Christopher G. Lowe. *Method Development for Implementation of Liver Biopsy Sampling to Quantify Organochlorines in Elasmobranchs* – Western Society of Naturalists 2024. Portland, Oregon. November 7-10, 2024.

## Grants and Scholarships

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**ARI Next Gen Scholarship (\$10,000) - August 2025**

- Exposure to current-use and legacy pesticides in sea urchins.

**Graduate Student Research Funds (\$400) - August 2025**

## Awards

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**President's Team Achievement Award: Shark Beach Safety Team**

- CSULB Shark Lab 2024

## Professional Services

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**SoCal SETAC Co-Student Board Representative (2025-2027)**

- Attend monthly board meetings in preparation for the annual conference, edit videos for newsletters and social media, and co-run the social media (LinkedIn, Instagram, and TikTok)

## Teaching and Outreach Experience

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**Teaching Associate, Cal Poly Pomona**

Fall 2025- Present

Bio 1220: Evolution, Ecology, and Biodiversity Lab

- Presented short lectures during class, set up and facilitated lab activities, and developed quizzes based on key concepts of Evolution, Ecology, and Biodiversity

**Shark Lab, CSULB**

August 2021-May 2025

Supervisor: Dr. Chris Lowe

- *Education Programs*- K-12 education programs geared toward elasmobranch biology and ecology
- *Lab Tours*- lead tours to inform the public on current research and native marine species
- *Shark Shacks*- outreach educational booth on research and technology
- *Junior Lifeguard Presentations*- present elasmobranch biology, research technology, and beach safety to junior lifeguards in Southern California

## Volunteer Experience

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**Shark Lab, CSULB**

August 2021-December 2023

Supervisor: Dr. Chris Lowe

- *Lab Skills*- stable isotope processing for muscle samples, eDNA water filtration
- *Biopsies*- retrieved muscle biopsies for stable isotope analysis
- *Data Retrieval*- downloaded data from underwater acoustic receivers and EnvLoggers
- *Data Analysis*- analyzed drone surveys spotting human/shark interactions, georeferenced drone surveys using ArcMap, and analyzed BRUV videos

**Aquarium of the Pacific, Long Beach**

April 2021-November 2021

Education Ambassador

- Educated the public on the species on exhibit in small and large group communication
- Monitored touch tanks

## Research Mentoring

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**Luci Herrera-Lopez** (CSULB Undergraduate 2024-2025)

**Brayden Ortiz** (CSULB Undergraduate 2024-2025)

## Professional Training

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- CITI Working with IACUC- Lab Animal Research
- CITI Working with Fish in a Research Setting
- Certified PADI Open Water Diver
- Official California Boaters Card
- CPR
- First aid
- Water Safety for Swim Instructors
- Heat Acclimatization and Heat Illness Prevention
- Concussion in Sports
- Lab Safety Fundamentals

- Fundamentals of Biosafety
- Biosafety Hazardous Waste Handling and Disposal

## Computer Skills

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### Intermediate

- Excel, Word, Python, R, ImageJ, and GC-MS analysis

### Beginner

- ArcGIS and ArcMap

## References

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Andrea Bonisoli-Alquati, CPP (Graduate Advisor)

[aalquati@cpp.edu](mailto:aalquati@cpp.edu)

Varenka Lorenzi, CSULB IIRMES

[Varenka.Lorenzi@csulb.edu](mailto:Varenka.Lorenzi@csulb.edu)

Christopher G. Lowe, CSULB Shark Lab

[chris.lowe@csulb.edu](mailto:chris.lowe@csulb.edu)

Kady Lyons, California Marine Sanctuary Foundation

[kady.lyons@sbcglobal.net](mailto:kady.lyons@sbcglobal.net)